VISUAL IMAGE INTERPRETATION
Understanding Image and Processing

IMAGE INTERPRETATION

- When we look at images, we see various objects of different shape and sizes
- Some of these may be readily identifiable while other may not be, depending upon individual's perceptions and experience
- When we can identify what we see on the images and communicate to others we are practicing IMAGE INTERPRETATION
- An act of examining images for the purpose of identifying objects and judging their significance

VISUAL IMAGE INTERPRETATION

- Most intuitive way to extract information from RS data based on the human's ability to relate colors, shapes and patterns in an image to the real world features
- Human vision goes a step beyond color perception, rather the ability of a person to draw conclusions by visual observation
- Training and experience with keen power of observation, imagination and patience
- Results depends upon
  - Nature of objects or phenomena
  - Quality of image
  - Other supporting materials such as:
    - Maps and reports
    - Field observations

ELEMENTS OF VISUAL INTERPRETATION

- Shape
- Tone
- Pattern
- Site
- Resolution
- Size
- Shadow
- Texture- (For a group of neighboring pixels)
- Association

Shape





Rectangular features often indicate human influence

Size





Pattern

Size and shape information greatly influenced by image resolution

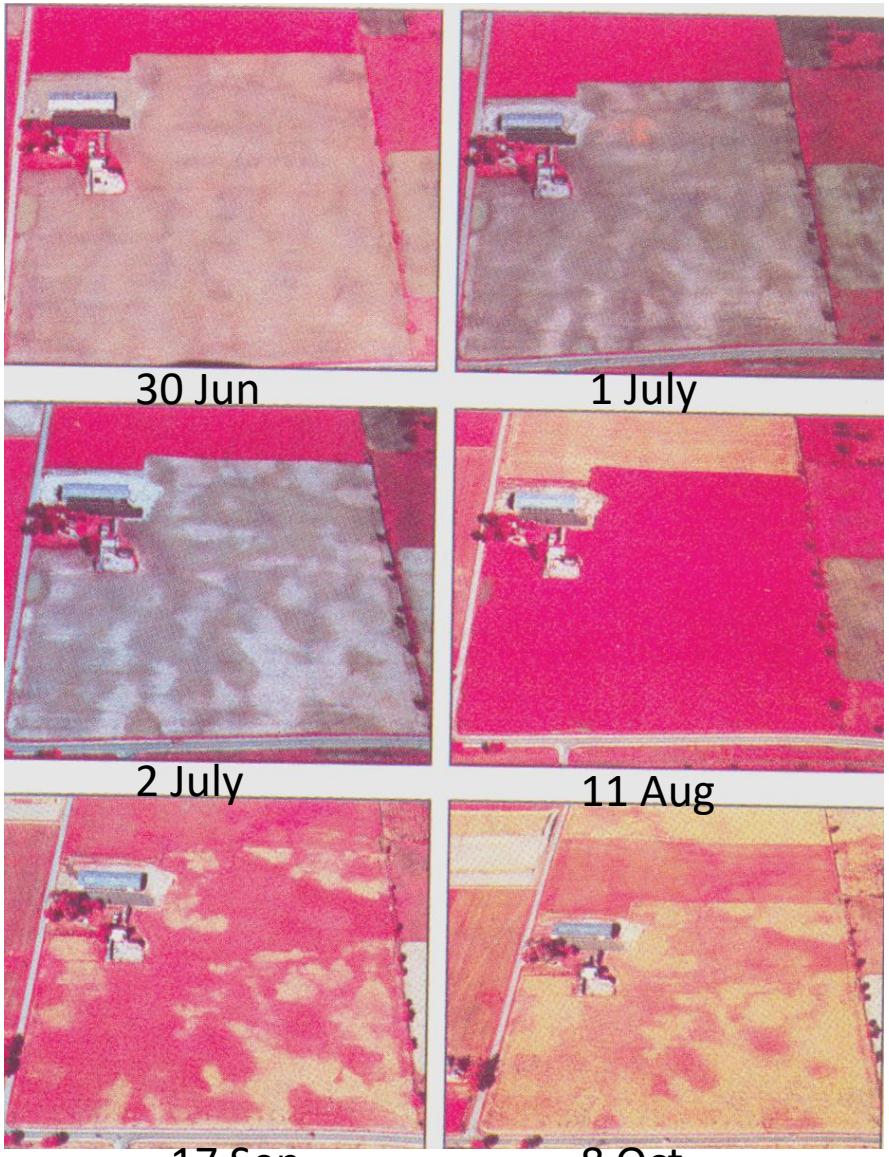
Texture





Shadow

Tone



SITE

- The location of a feature narrows the list of possible cover types
- Site refers to topographic and geographic location of the objects of study
- Certain type of vegetation or tree can only occur in certain geographic area not every where
- Relative location particularly useful to determine land use
- A large building near a converging network of railroads should be railway station, not a hospital

ASSOCIATON

- Relationship between other recognizable objects or features in the proximity of the target of interest
- Identification of features that one would expect to associate with other features may provide information to facilitate identification
- Commercial properties may be associated with proximity of major transportation routes and fairly large parking areas, whereas residential areas with schools, playgrounds

RESOLUTION

- **It depends upon many factors but always plays an important role for identification**
- **Knowing the resolution of the image helps to convert feature dimensions on the image to actual dimensions**
- **Some object are too small or have too little contrast with their surroundings to be clearly seen on the image**

STRATEGIES

- Image interpretation process can involve various levels of complexity, from simple direct recognition of object in the scene to inference of site conditions
- Direct recognition-identification of highway interchange
- On the other hand, it may be necessary to infer rather than directly observe the characteristics of features based on their appearance on images
- Conditions of site: Moist soils, backfill lose soil, can help infer the presence of buried objects
- Knowledge of the crop calendar for an area would help crop identification, when the crop be visible, particular growing

STRATEGIES

- In a sense, image interpretation is the work of a detective trying to put all pieces of information and evidence together to solve the case
- For interpreter, the mystery might be that why certain area in agricultural field looks different from the rest of the field

Reference/Readings

- Chapters on Image Classification

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